

Scalability & Future Plan

Date	15 March 2024
Team ID	SWUID2026317864
Project Name	Farm Management System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Requires an internet connection; limited offline access.	Farmers in rural or low-connectivity areas cannot enter or view farm and crop records in the field, which can cause delayed or missed updates.	High
2	Dependency on Salesforce licenses; the Developer Org allows only 2 Salesforce-licensed users at a time.	Restricts the number of active farmers, managers and buyers who can use the system, and adds licensing cost when the user base grows.	High
3	Basic automation and analytics: only validation rules, standard reports and dashboards; no predictive or external data support.	Managers must interpret trends manually; no crop-yield prediction, weather or market-price insights to support decisions.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Developer Org with limited licenses; users are the System Admin, Farm Manager and Sales Department, managed through profiles and roles.	Move to a production Salesforce edition with additional Platform licenses; add Experience Cloud (community) portal access for farmers and buyers; extend the role hierarchy and sharing rules for new regions.
2	Data Storage	Four custom objects (Farm, Crop, Farmer, Buyer) with relationships stored on Salesforce Cloud within Developer Org storage limits.	Upgrade storage capacity; archive old crop seasons using Big Objects or scheduled data export; add indexed fields and regular data clean-up as multiple farms and regions are added.
3	Performance	Standard reports and dashboards with a target of reports in under 3 seconds; tested for smooth response with multiple user operations.	Use filtered and indexed report criteria, roll-up summary fields, and scheduled dashboard refresh; optimize flows and run load testing in a sandbox before releasing changes.
4	Security	Role-based access through profiles (Farm Manager, Sales Department), roles, sharing rules (Crop to Farm Manager, Buyer to Sales) and validation rules.	Enable multi-factor authentication, field-level security and permission sets, login IP and hours restrictions, encryption for sensitive data, and periodic access audits.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Mobile application integration (Salesforce Mobile App) with offline data capture for farmers.	0 - 3 months	Field-level data entry without constant connectivity; fewer delays and errors.
Phase 2	Weather data integration and IoT sensor connectivity (soil moisture, temperature).	3 - 6 months	Real-time field conditions on dashboards for better irrigation and crop planning.
Phase 3	AI-based crop prediction and market price forecasting.	6 - 12 months	Improved yield planning and better selling decisions for farmers and buyers.
Phase 4	Government scheme integration and multi-region rollout with farmer and buyer community portal.	12 - 18 months	Wider adoption across farms and regions; easier access to subsidies and schemes.